Задание 3

Построение филогенетических деревьев

Найдём лучшую модель молекулярной эволюции для ML дерева.

Results

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Table. Maximum Likelihood analysis of substitution models

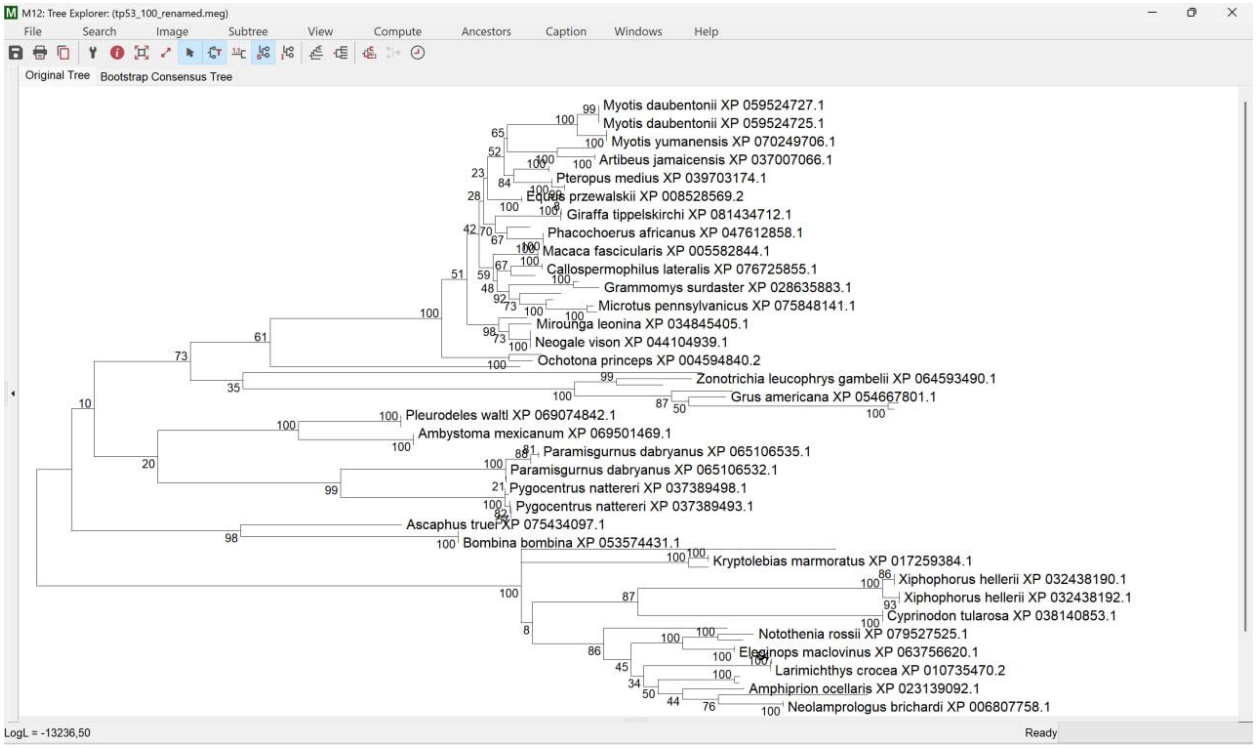
Model	Parameters	BIC	AICc	*lnL*	(+*I*)	(+*G*)	*f*(A)	*f*(R)	*f*(N)
JTT+G+I	199	28542,539	26881,393	-13240,428	0,12	0,88	0,077	0,051	0,043
JTT+G+I+F	218	28611,942	26792,460	-13176,708	0,12	0,86	0,065	0,075	0,044
WAG+G+I	199	28782,607	27121,461	-13360,462	0,12	0,90	0,087	0,044	0,039
WAG+G+I+F	218	28879,492	27060,010	-13310,482	0,12	0,89	0,065	0,075	0,044
Dayhoff+G+I	199	28917,345	27256,199	-13427,831	0,12	0,90	0,087	0,041	0,040
Dayhoff+G+I+F	218	28939,338	27119,856	-13340,405	0,12	0,85	0,065	0,075	0,044
LG+G+I+F	218	28951,252	27131,769	-13346,362	0,12	0,83	0,065	0,075	0,044
LG+G+I	199	28955,943	27294,797	-13447,130	0,12	0,82	0,079	0,056	0,042
rtREV+G+I+F	218	29086,859	27267,377	-13414,166	0,12	0,84	0,065	0,075	0,044
cpREV+G+I+F	218	29175,077	27355,595	-13458,275	0,12	0,78	0,065	0,075	0,044
rtREV+G+I	199	29183,504	27522,358	-13560,910	0,12	0,85	0,065	0,045	0,038
cpREV+G+I	199	29196,720	27535,574	-13567,519	0,12	0,83	0,076	0,062	0,041

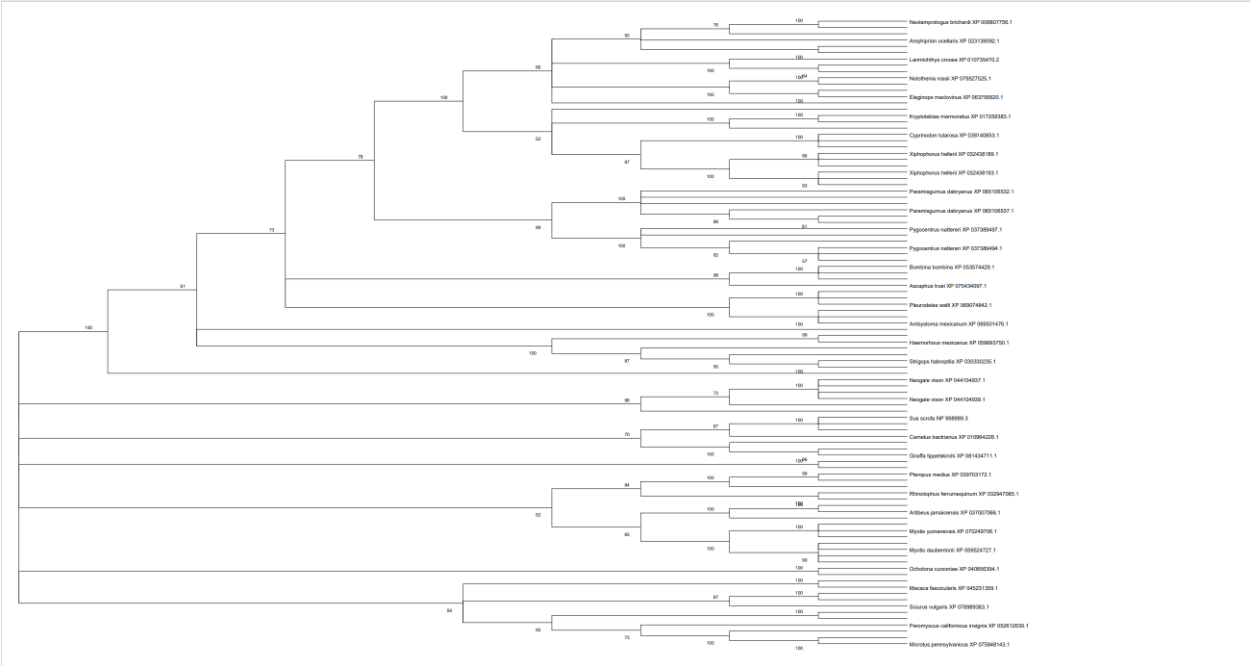
Для построения ML-дерева была найдена лучшая модель методом Best-Fit Substitution Model (Maximum Likelihood). Лучшая модель: JTT + G + I (BIC = 28542,539 — минимальное значение среди всех протестированных моделей).

Построим ML-дерево (Maximum Likelihood) с использованием модели JTT + G + I и параметром bootstrap = 100.

Likelihood Phylogeny Reconstruction

Option	Setting
ANALYSIS	
Statistical Method →	*Maximum Likelihood*
PHYLOGENY TEST	
Test of Phylogeny →	*Standard Bootstrap (slow)*
Bootstrap Replicates →	*100*
SUBSTITUTION MODEL	
Substitutions Type →	*Amino acid*
Model/Method →	*Jones-Taylor-Thornton (JTT) model*
RATES AND PATTERNS	
Rates among Sites →	*Gamma Distributed With Invariant Sites (G+I)*
No of Discrete Gamma Categories →	*5*
DATA SUBSET TO USE	
Gaps/Missing Data →	*Use all sites*
TREE INFERENCE OPTIONS	
ML Heuristic Method →	*Nearest-Neighbor-Interchange (NNI)*
Initial Tree for ML →	*Make initial tree automatically (Default - NJ/MP)*
Branch Swap Filter →	*None*
SYSTEM RESOURCE USAGE	
Number of Threads →	*16*





Построим NJ-дерево (Neighbor-Joining) с моделью JTT + G и параметром bootstrap = 100.

M12: Analysis Preferences

NJ Phylogeny Reconstruction

Option	Setting
ANALYSIS	
Scope	*All Selected Taxa*
Statistical Method →	*Neighbor-joining*
PHYLOGENY TEST	
Test of Phylogeny →	*Bootstrap method*
Bootstrap Replicates →	*100*
SUBSTITUTION MODEL	
Substitutions Type →	*Amino acid*
Model/Method →	*Jones-Taylor-Thornton (JTT) model*
RATES AND PATTERNS	
Rates among Sites →	*Gamma Distributed (G)*
Gamma Parameter →	*1,00*
Pattern among Lineages →	*Same (Homogeneous)*
DATA SUBSET TO USE	
Gaps/Missing Data →	*Partial deletion*
Site Coverage Cutoff (%) →	*95*
SYSTEM RESOURCE USAGE	
Number of Threads →	16

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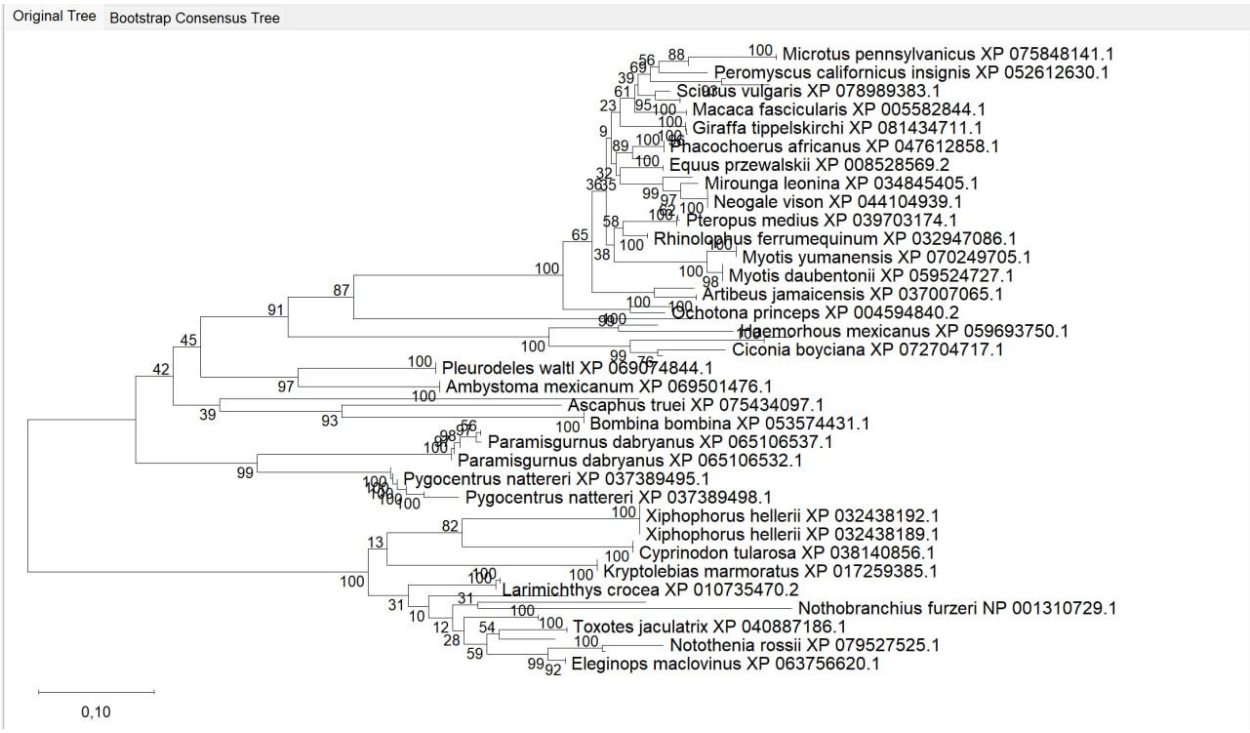


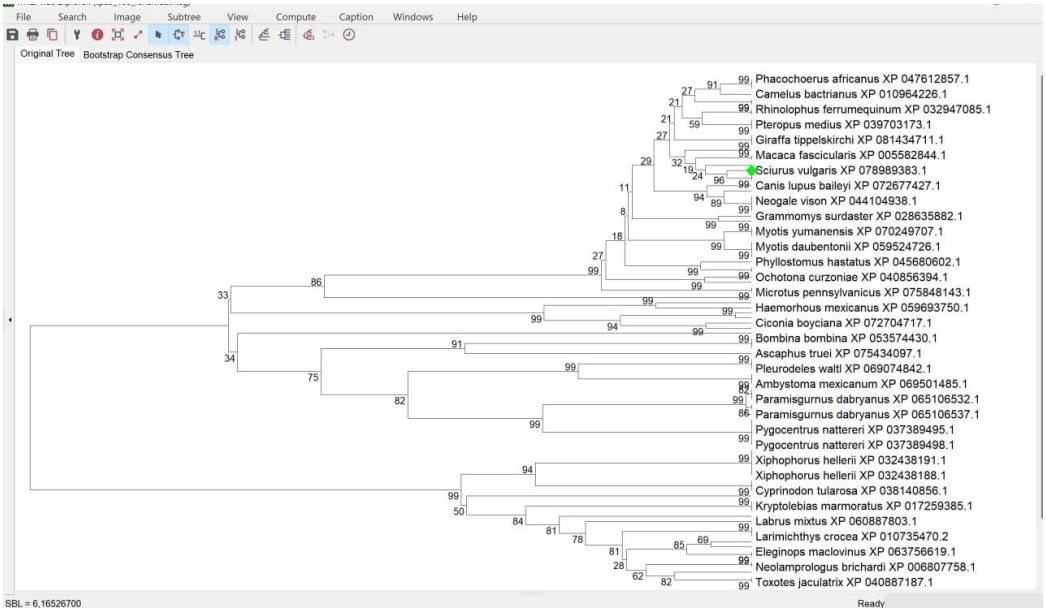
Построим UPGMA-дерево с моделью JTT + G и параметром bootstrap = 100.

M12: Analysis Preferences

UPGMA Phylogeny Reconstruction

Option	Setting
ANALYSIS	
Scope	*All Selected Taxa*
Statistical Method →	*UPGMA*
PHYLOGENY TEST	
Test of Phylogeny →	*Bootstrap method*
Bootstrap Replicates →	*100*
SUBSTITUTION MODEL	
Substitutions Type →	*Amino acid*
Model/Method →	*Jones-Taylor-Thornton (JTT) model*
RATES AND PATTERNS	
Rates among Sites →	*Gamma Distributed (G)*
Gamma Parameter →	*1,00*
Pattern among Lineages →	*Same (Homogeneous)*
DATA SUBSET TO USE	
Gaps/Missing Data →	*Partial deletion*
Site Coverage Cutoff (%) →	*95*
SYSTEM RESOURCE USAGE	
Number of Threads →	*16*

Help | Cancel | Ok



PDF-файлы с деревьями прикреплены в папку с дз на GitHub.

Все три дерева построены на одном и том же выровненном файле tp53_100_renamed.meg (100 ортологов белка TP53).